How Order Holds

On Active Boundaries, Self-Maintaining Systems, and the Cost of Staying Organized

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Cripsis Logic

An essay introducing the academic paper that follows.

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Why does anything stay together? Not metaphorically — literally. A living cell is mostly water, salt, and a tangle of long-chain molecules that have no particular reason to remain organized. Left to its own devices, the universe pulls things apart. The second law of thermodynamics is uncompromising about this: in any closed system, disorder increases. And yet, for a few decades at a time, that bag of salty water stays organized enough to read this sentence. So does a redwood. So does a coral reef, a city, an economy, the magnetic field of a star, the routing system that delivered this essay to your screen. How? The honest answer, the one physicists have been circling since the nineteenth century, is that none of these things are closed systems. They each maintain their structure by doing active work — taking in energy or information from outside, exporting disorder back to the environment, and burning the difference to hold their shape. A cell metabolizes. A redwood photosynthesizes. A city imports food and exports trash. A star fuses hydrogen against gravitational collapse. A computer system handling a million requests per second runs algorithms that decide what to admit, what to defer, and what to reject, all in service of not falling over. These examples sound like very different problems. They are. The machinery a cell uses to stay alive bears little resemblance to the machinery a star uses to stay lit, which bears little resemblance to the machinery a software system uses to stay responsive. The microscopic mechanisms diverge wildly. But at a higher level of description — the level a physicist would call phenomenological, meaning concerned with patterns of behavior rather than with underlying mechanism — these systems share something striking. They all exhibit thresholds. They all have regimes where they hold their shape and regimes where they don't. They all, when pushed too far, collapse suddenly rather than gradually. And the line between holding and collapsing tends to be sharper than intuition would suggest. This essay is about that line, and about a small contribution to understanding where it lives. It will not assume any background in physics or mathematics. It will assume a reader who finds the question interesting and is willing to spend an hour thinking it through.

I. The Long Conversation

The question of how order persists in systems far from equilibrium has a long pedigree. It is, in some sense, the central question of twentieth-century physics outside the relativity-and-quantum mainstream — the question that occupied the people who looked at the universe and asked not what it is made of, but how it holds itself up. Erwin Schrödinger, the physicist who gave us the wave equation that underwrites quantum mechanics, spent the last years of his life writing a small book called What Is Life? In it, he argued that living things stay alive by feeding on what he called negative entropy — by extracting order from their environment to compensate for the disorder they generate internally. The book is brief, speculative, and occasionally wrong, but it set the agenda for a generation of biophysicists and gave us the conceptual vocabulary we still use. Schrödinger's central insight was that life is not a violation of thermodynamics but an expression of it: living systems are local regions of decreasing entropy, paid for by larger increases in entropy elsewhere. The cost is real. The accounting balances. Ilya Prigogine, working in Brussels a generation later, formalized this into the mathematics of dissipative structures. Prigogine showed that when energy flows through a system far enough from equilibrium, ordered patterns can emerge spontaneously — convection cells in a heated fluid, chemical clocks that oscillate between states, the Belousov–Zhabotinsky reaction with its hypnotic spiral waves. Prigogine won the Nobel Prize in 1977 for this work. His central message was that order is not the opposite of chaos; under the right conditions, order is what chaos produces. Stuart Kauffman, at the Santa Fe Institute, took this further. In a series of provocative books beginning in the late 1980s, Kauffman argued that life sits at what he called the edge of chaos — a regime where systems are organized enough to function but disordered enough to adapt. Too much order produces frozen, brittle systems unable to respond to change. Too much chaos produces noise. The interesting regime, Kauffman claimed, is the narrow band between, and evolution drives systems toward it. Whether this is literally true remains debated. That it is suggestive remains undeniable. More recently, Karl Friston, a British neuroscientist, has framed perception and cognition themselves as a special case of this larger question. Brains, in Friston's account, stay organized by minimizing the difference between what they predict about the world and what they observe. This minimization is itself an active process — the brain feeds on information the way cells feed on glucose, and the suppression of surprise is metabolically expensive. Friston's free-energy principle is controversial, dense, and possibly the most ambitious attempt yet to unify thermodynamics, information theory, and biology under a single mathematical roof. Around these named figures has grown a broader tradition — sometimes called complexity science, sometimes complex adaptive systems, sometimes simply the Santa Fe school — that has pursued the question across biology, economics, ecology, and computation. Researchers in this tradition have produced a generation of insights about emergence, self-organization, adaptive complexity, networks, and the strange regularities that appear when many simple agents interact under constraint. Three more specific research programs deserve mention because the paper that follows engages with them directly. Friston and his collaborators have developed the formalism of Markov blankets — statistical boundaries that separate a system's internal states from its environment — as the mathematical object on which active inference operates. The viability theory tradition, beginning with Jean-Pierre Aubin's 1991 monograph, formalized self-maintenance through the geometry of viability kernels in control systems. And in systems biology, James Sethna, Ilya Nemenman, and their collaborators have shown that complex adaptive networks often reduce, through manifold-boundary approximations of their microscopic kinetics, to a small number of dimensionless ratios — sometimes just one. Each of these programs answers a version of our central question with different machinery. Each is a serious body of work. None of them quite produces the kind of compact, falsifiable phenomenological law this essay is leading toward, though all of them frame what such a law would have to do. Each of these framings captured part of the phenomenon. Each of them fell short of the same thing: a phenomenological law compact enough to be tested against measurement in a specific system, then extended only as the data warranted. The frameworks were rich. The vocabulary was evocative. The predictions were rare. This is not a complaint, exactly. The work was hard, the systems were varied, and the temptation to generalize was strong. But popular science writing about complex systems has long had a problem: it tends to substitute metaphor for prediction. A theory that explains why an ant colony, a financial market, and a human brain are all examples of the same deep principle is exciting to read about and difficult to test. A theory that says here is the specific number you should measure, and here is what you should see if I am right, and here is what you should see if I am wrong is much less exciting and much more useful. What follows is an attempt at the second kind.

II. A Question Worth Asking Precisely

Here is the question, sharpened. Take any system that does active work to maintain its organization — a cell, a neural circuit, a software service, a market-making operation, a small team trying to ship a product. Suppose we leave that system alone. We don't add new energy, we don't add new information, we don't intervene from outside. Will the system sustain its organized state, or will it collapse to something inert? Phrased this way, the question admits a sharp answer for a surprisingly broad class of systems. The answer is that there exists, for each system, a single dimensionless number — a ratio of how strongly the system feeds back on itself versus how strongly it dissipates internal complexity — that determines viability. Below a critical value, the system cannot hold its organization. Above the critical value, it can. The transition between these regimes is not gradual. It is a threshold. This may sound similar to a number readers will have heard about during the recent pandemic: the basic reproduction number, R-zero, which determines whether an infectious disease will spread or die out. The structural similarity is real. Both numbers are dimensionless ratios that cross one as the qualitative behavior of the system changes. Both correspond to thresholds in the underlying mathematics. But the mechanism is entirely different. R-zero is about transmission between hosts. The number that governs the question we're asking here — call it R, plain — is about a single system's ability to feed back on itself faster than it dissipates the complexity its own activity produces. The two numbers share a logical shape, not a physical origin. Why should such a number exist? Because the mathematics of self-maintenance, when written carefully, has a specific structure. Imagine a system with two coupled quantities. The first is how much organized, useful activity the system has at any given moment — call this the activity. The second is how much disorder, complexity, or internal pressure has built up — call this the entropy, in a deliberately broad sense. These two quantities feed each other. Activity generates internal pressure, because every action produces consequences that have to be processed. Pressure drives further activity, because organized work is often a response to internal complexity that demands attention. But both quantities also decay over time if nothing replenishes them. Activity dissipates without ongoing drive. Pressure relaxes if there's nothing generating new of it. The question of whether the system can sustain itself is the question of whether this two-way coupling is strong enough to overcome the dissipation. If the feedback is weak relative to the decay, the system winds down to nothing. If the feedback is strong relative to the decay, the system can hold a stable, bounded level of activity indefinitely. The viability number R captures this comparison in a single quantity. If R is less than one, dissipation wins. If R is greater than one, feedback wins, and a stable active state becomes available. There is one more piece of mathematical machinery the model needs. Without it, the equations would predict that activity grows without bound — runaway feedback, infinite escalation, the kind of behavior that real systems do not exhibit because real systems have limits. Cells run out of substrate. Software systems hit hard rate caps. Stars exhaust their fuel. To capture this, the model includes a saturation term: a nonlinear effect that becomes important only when activity gets large, and that bends the trajectory back from runaway. The specific form of this saturation determines exactly how the active state scales with how far above threshold the system is, but the qualitative story is the same regardless: above threshold, the system settles into a stable, bounded active state, and the size of that state grows in a specific predicted way as you push the system further from the threshold.

III. The Shape of the Answer

When this is all written down carefully, three predictions fall out, and they are predictions you could in principle test against any real system you suspect the framework applies to. The first prediction is the threshold itself. Sweep R across one — by tuning whatever parameter in the system controls the feedback — and the system should transition sharply from quiescent to active. Below the threshold: no spontaneous activity. Above it: bounded, stable activity. The transition should be visible without elaborate statistical machinery, simply by plotting the system's organized activity against the value of R. The second prediction is the amplitude scaling. Above the threshold, the size of the active state should grow as the square root of how far you are above one. Equivalently, the square of the activity should grow linearly with distance above threshold. This is a quantitative claim. It is testable. A system that crosses the threshold but doesn't show this specific scaling has falsified the model — or at least the specific saturation form the model assumes. The third prediction is the most distinctive, and the one that connects this work to a phenomenon readers may have encountered in popular science before. It is called critical slowing, and it works like this. Near the threshold, when the system is just barely able to maintain itself, its recovery from disturbance becomes arbitrarily slow. Push the system a little — small perturbation, nothing dramatic — and watch how long it takes to return to its stable state. Far above threshold, it returns quickly. Just above threshold, it returns slowly. Right at threshold, it doesn't return at all in any practical sense; the recovery time diverges to infinity. This is not a curiosity. Critical slowing has been observed in ecosystems heading toward collapse, in lakes transitioning between clear and turbid states, in financial markets in the months before crashes, and in human brains transitioning between states of consciousness or between healthy and seizure activity. It is one of the more reliable signatures we have that a system is approaching a qualitative transition. The framework here predicts it from first principles, with a specific functional form: the recovery time should grow as one over the distance above threshold. This is a sharp, checkable claim. These three predictions — the threshold, the amplitude scaling, the critical slowing — are what the framework commits to. If a calibrated real system exhibits all three with the right functional forms, the framework fits that system. If it exhibits none of them, the framework is wrong. If it exhibits some but not others, the failure modes tell us specifically what kind of extension the framework needs — different saturation, additional state variables, delayed feedback, and so on. The framework is, in other words, falsifiable in a precise way, and informative even when it fails.

IV. The Discipline of Being Wrong

It is worth dwelling on this last point, because it is the part of the work I think matters most, and it is the part most likely to be underappreciated. The history of cross-disciplinary phenomenological theory is littered with frameworks that explained too much. They were elegant, evocative, often profound — and they accommodated almost any observation, which is to say, they predicted almost nothing. The philosopher Karl Popper argued, famously, that a theory which cannot in principle be falsified is not a scientific theory at all. He was thinking primarily of psychoanalysis and Marxism, but the warning applies more broadly. Theories of complexity, in particular, have a tendency to drift toward the unfalsifiable. They explain why systems are complex by appealing to complexity. They predict that systems will self-organize, and when systems do something else, that too is self-organization. The way to do better is to commit, in advance, to specific things you would not see if you were right, and to admit clearly when you see them. The framework presented here does this. It includes an explicit table of failure modes — observations that, if seen in a calibrated system, would either falsify the model or tell us exactly what kind of extension is needed. Sustained activity below threshold? Falsified, unless we missed external forcing or a hidden state variable. Persistent oscillations around the active state? The minimal model is incomplete and needs delayed feedback or an additional state. No critical slowing near threshold? Either we mis-estimated R, or the transition isn't the one the model describes. This kind of explicit failure-mode mapping is unusual in popular treatments of complexity science, and it is the part of the paper I would point a skeptical reader to first. The mathematics is well-established, drawn from a long tradition of nonlinear dynamics. The interpretive contribution is genuine but modest. The discipline of being clear about what would prove the framework wrong is, I think, the part that earns the rest of it. A framework like this becomes scientifically useful only if it can be wrong. The framework presented here can be wrong in six specific ways. Each way of being wrong tells us something. That is the whole structure.

V. Where This Touches the World

The most immediate application is in software systems that regulate their own behavior under load. This sounds dry, but it shouldn't. As we deploy artificial intelligence at scales that strain the systems carrying it, the question of whether our control infrastructure is doing real regulatory work or merely adding overhead becomes economically and operationally consequential. Concretely: when you send a request to a large AI model, that request doesn't go directly to the model. It passes through a series of regulatory layers — admission controllers, routing logic, backpressure systems — that decide whether to serve your request immediately, queue it, defer it, or reject it. These regulatory layers are themselves active boundaries, in the precise sense the framework describes. They take in workload, generate internal complexity in the form of queue pressure and decision uncertainty, and dissipate that complexity through completion, retries, and cleanup. They have all the structural features of the systems the framework was designed to describe. They also produce, as a routine matter of operation, the telemetry needed to test the framework's predictions. Queue depth, decision rates, recovery times — these are logged in nearly every production system at scale. The framework gives operators of such systems a quantitative answer to a question they currently answer with intuition: is our control layer actually doing useful work? Is it stabilizing the system, or is it merely adding overhead without protective value? Given the cost of running large AI models — measured in millions of dollars per month for any serious deployment — the difference between a control layer that genuinely regulates and one that merely consumes is substantial. Beyond this immediate application, the framework plausibly applies wherever the question "is this system maintaining itself, or is it being maintained?" is meaningful. Neural circuits operate near criticality, and the framework's predictions about critical slowing and amplitude scaling have natural neural analogs. Ecological systems exhibit threshold transitions and recovery-time signatures that the framework predicts. Organizations and institutions undergo discontinuous transitions in operational coherence that look, structurally, like the active-to-quiescent transitions the model describes. Markets, particularly market microstructure and clearing infrastructure, regulate organized trading activity against pressure from order flow. Each of these would require its own calibration work, its own choice of proxies for activity and complexity, its own validation. The paper is explicit that these are research directions, not completed claims. The temptation to gesture broadly at universal applicability has been the downfall of many complexity frameworks, and this one tries to avoid the trap by naming specifically where the work has been done — synthetic demonstrations of the predicted signatures in software-style telemetry — and where it hasn't. There is also a longer-range direction worth flagging, though it lies outside the present work and is mentioned only as a structural resonance. The active-boundary description — a boundary that retains information while exchanging entropy with its environment under regulatory constraint — has a formal echo in black-hole horizon physics. Black-hole horizons are themselves active boundaries in this sense: they retain information about what fell in, exchange entropy with their cosmic environment via Hawking radiation, and obey laws that look thermodynamic. Whether the framework here has anything to say about gravitational systems is a question for a different paper and a different set of derivations. But the structural similarity is real enough to flag, and curious enough to be worth flagging.

VI. What the Work Doesn't Do

It is worth being explicit about the limits of what the framework claims, because those limits are part of why the work is offered in the spirit it is. It is not a microscopic theory. The framework does not derive its equations from underlying mechanisms in any specific domain. It is phenomenological — a description of patterns of behavior — and its parameters must be measured or estimated for each system to which it is applied. The viability number R is a ratio of quantities that have specific meanings in any given system, but those meanings have to be supplied by the practitioner, not by the framework. It is not a universal theory of complexity. It applies to systems where the active-boundary description fits. There are surely systems for which it does not fit — systems with strong oscillatory dynamics the minimal model cannot produce, systems with multiple stable states, systems where the relevant nonlinearity is not the cubic saturation the framework assumes. The paper is explicit about these cases and provides the failure modes that would diagnose them. It is not a substitute for empirical work in any specific domain. Calibrating the framework to a real cell, a real ecosystem, a real neural circuit, or a real organization is a research program, not a consequence of having read this paper. The synthetic demonstrations in the paper show what the predicted signatures look like in telemetry-shaped data, but synthetic demonstrations are not real calibrations and the paper does not claim otherwise. And it is not, despite its structural resonances with much grander questions, a theory of life or consciousness or markets or civilization. It is a small mathematical claim about a class of systems that exhibit a specific kind of self-maintenance, with a specific viability threshold and a specific set of falsifiable signatures. It is offered as one node in a long conversation, not as the conversation's conclusion.

VII. The Invitation

Why publish a paper like this? Why write an essay introducing it to readers who are unlikely to read the technical version? The honest answer is that work like this has value only if it circulates. A phenomenological framework gains traction when practitioners try it on their own systems. A model becomes useful when other people calibrate it, extend it, falsify it, or replace it with something better. None of that happens if the paper sits on a preprint server unread. More than that, the question the framework addresses is one that matters. We are entering a period where the systems we build are increasingly active boundaries in their own right. Large AI models regulate their own load. Distributed systems negotiate their own state. Autonomous agents make their own decisions about when to act and when to wait. The discipline of asking, of any such system, whether it can sustain itself and at what cost, is going to become more important rather than less. The vocabulary for asking that question precisely has been developing for a century, and this work tries to add to it. If you are a researcher in nonlinear dynamics, complexity science, or computational systems and you find the framework interesting, the technical paper that follows lays out the mathematics, the stability analysis, the numerical verification, and the falsification criteria in full. The reproducibility archive is public; the equations and predictions are testable on any system that exposes the right operational variables. If you are a reader who came to this essay because the question interested you, and the technical paper that follows is more than you want to read in one sitting, that is fine. The essay you have just read is meant to stand on its own. The paper is here for those who want it. Either way, the conversation is older than this contribution and will continue past it. Schrödinger asked what life was. Prigogine asked how order emerged. Kauffman asked where adaptation lived. Friston asked how brains stayed organized. Each of them got something, missed something, and handed the question forward. This paper is an attempt to pass it forward in turn, with one specific contribution and one explicit set of ways it could be wrong. Order holds because something pays the bill. The question is always what the bill is and whether the system can afford it. That question is worth asking precisely. The paper that follows is one attempt at doing so.

Coda

The technical paper begins on the next page. It is a short paper — sixteen pages, with five figures, a falsification table, and two appendices — and it is written in the spare register that academic physics requires. Readers who have followed the essay this far should find the equations more transparent than they might expect, because the variables and parameters have already been introduced in plain English. The mathematics is doing the work of making the claims precise; the essay has done the work of explaining why the claims matter. If you read only the abstract and the falsification table, you will have the structural argument. If you read the full paper, you will have the proof. If you read neither, you will still, I hope, leave with something — a clearer sense of how a question about why anything stays together becomes a specific mathematical claim, and what it means to make such a claim in a way that can be checked. Thank you for reading.

The Adaptive Boundary Equation:

A Minimal Active-Boundary Model for Self-Organizing Information–Entropy Dynamics

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Abstract

Adaptive systems persist by regulating information-bearing activity and entropy or complexity across boundaries. This paper proposes the Adaptive Boundary Equation (ABE), a normalized two-variable phenomenological model

$$\frac{dI}{dt} = \gamma S - \lambda I - \beta I^3, \qquad \frac{dS}{dt} = \eta I - \xi S,$$

with positive parameters. The cubic saturation term is the minimal nonlinearity required to bound the active state. The model yields a dimensionless viability number

$$R = \frac{\gamma\eta}{\xi\lambda}.$$

The quiescent boundary is stable for $R < 1$. For $R > 1$, the quiescent state loses stability and a stable positive active branch appears:

$$I^* = \sqrt{\frac{\gamma\eta/\xi - \lambda}{\beta}}, \qquad S^* = \frac{\eta}{\xi} I^*.$$

Linear stability analysis and numerical verification confirm the threshold, active-branch stability, amplitude relation, and critical-slowing behavior near the viability transition. A worked synthetic operational demonstration in the style of an adaptive computational governor reproduces the predicted threshold transition and amplitude scaling. The contribution is not a new dynamical-system normal form, but a compact adaptive-boundary interpretation of familiar nonlinear machinery, together with explicit falsification criteria and one synthetic operational demonstration.

Keywords: adaptive systems; active boundaries; information dynamics; entropy production; self-organization; viability threshold; critical transitions; point processes; nonlinear dynamics; phenomenological models; active inference

1 Introduction

The central problem of adaptive systems is not merely how order appears, but how order persists under flux [1]. Biological cells, neural circuits, institutions, markets, and computational governors remain viable only by regulating what crosses their boundaries, what is retained, what is dissipated, and what is forgotten. Each domain has its own microscopic mechanisms. Yet at the phenomenological level, many such systems exhibit similar signatures: threshold behavior, bounded active states, relaxation after perturbation, and sometimes damped oscillatory recovery.

Existing frameworks capture parts of this problem. Dissipative-structure theory [2, 3] explains the emergence of order far from equilibrium. Maximum-entropy-production arguments [4, 5] emphasize irreversible throughput. Free-energy formulations [6, 7] frame adaptation as inference under uncertainty. Control theory studies stabilization under feedback. Nonlinear dynamics supplies a mature language for thresholds, saturation, oscillation, and bifurcation.

The purpose of this paper is deliberately narrow. It does not claim a microscopic derivation of biological, economic, gravitational, or computational adaptation. It proposes a minimal active-boundary equation whose variables are interpreted as information-bearing activity and internal entropy or complexity. The paper then derives a viability threshold and shows what signatures would have to appear if the model is applicable to a given system.

This discipline is important. A model of this kind becomes scientifically useful only if it can be wrong. The ABE is therefore presented as a compact phenomenological target: simulate it, fit it, test whether the predicted threshold and amplitude scaling appear, and reject it where they do not.

2 Variables and physical interpretation

The model tracks two state variables.

- $I(t)$: *active information amplitude.* This is the boundary's organized information-bearing state, measured in a domain-specific unit such as effective memory strength, encoded state density, policy-bearing information, or normalized adaptive activity.
- $S(t)$: *internal entropy or complexity.* This is the boundary's internal microstate diversity, disorder, adaptive complexity, or pressure generated by information processing and dissipated by regulation, compression, thermalization, or forgetting.

The normalized model uses five positive parameters.

- γ: coupling gain from internal entropy or complexity into information-bearing activity.
- λ: regulatory decay of active information.
- β: nonlinear saturation coefficient limiting runaway activity.
- η: entropy or complexity generation rate per unit active information.
- ξ: entropy or complexity dissipation rate.

An optional forcing term $\Phi(t)$ represents external boundary drive. For threshold analysis $\Phi(t)$ is set to zero so that the question is endogenous viability: can the boundary maintain an active information-bearing state without continuous external forcing?

3 The normalized Adaptive Boundary Equation

The normalized active-boundary form of the ABE is

$$\frac{dI}{dt} = \gamma S - \lambda I - \beta I^3 + \Phi(t), \tag{1}$$

$$\frac{dS}{dt} = \eta I - \xi S. \tag{2}$$

Equation (1) says that information-bearing boundary activity grows when internal entropy or complexity provides useful adaptive drive, decays under regulatory loss, and saturates when activity becomes too large to remain linear. Equation (2) says that information-bearing activity generates entropy or complexity, while entropy or complexity dissipates at rate ξ.

The normalization removes an unnecessary response coefficient that would otherwise create dimensional ambiguity without changing the threshold. Time scaling or gain normalization can restore a domain-specific response rate if required. In this paper, the leading coefficient of the I equation is absorbed into the definitions of γ, λ, β, and Φ.

4 Viability threshold

Set $\Phi(t) = 0$ and solve for equilibria. From Eq. (2),

$$S^* = \frac{\eta}{\xi} I^*. \tag{3}$$

Substituting Eq. (3) into Eq. (1) gives

$$0 = \left(\frac{\gamma\eta}{\xi} - \lambda\right) I^* - \beta (I^*)^3. \tag{4}$$

The equilibria are

$$I^* = 0, \qquad S^* = 0, \tag{5}$$

$$I^* = \pm\sqrt{\frac{\gamma\eta/\xi - \lambda}{\beta}}, \qquad S^* = \frac{\eta}{\xi} I^*. \tag{6}$$

The physically relevant positive active branch exists only when $\gamma\eta/\xi - \lambda > 0$, equivalently

$$\eta > \frac{\xi\lambda}{\gamma}. \tag{7}$$

The negative branch is the signed counterpart of the normal form; in applications where I represents a nonnegative magnitude, the physical state is interpreted through $|I|$ or restricted to the positive branch. Define the dimensionless viability number

$$R = \frac{\gamma\eta}{\xi\lambda}. \tag{8}$$

Then $R > 1$ is the self-maintaining regime. The threshold is not imposed by interpretation; it follows directly from the equilibrium condition once the minimal saturation term is included. Below threshold, the only unforced equilibrium is quiescence. Above threshold, a bounded active branch becomes available.

5 Linear stability analysis

5.1 Quiescent branch

Linearizing Eqs. (1)–(2) at $(I, S) = (0, 0)$ with $\Phi(t) = 0$ gives

$$J_0 = \begin{pmatrix} -\lambda & \gamma \\ \eta & -\xi \end{pmatrix}. \tag{9}$$

The trace and determinant are

$$\mathrm{Tr}(J_0) = -(\lambda + \xi) < 0, \tag{10}$$

$$\det(J_0) = \lambda\xi - \gamma\eta = \lambda\xi(1 - R). \tag{11}$$

For $R < 1$, the determinant is positive and the trace is negative, so the quiescent state is stable. For $R > 1$, the determinant is negative, so the quiescent state is a saddle. In that regime, the quiescent boundary cannot be the stable self-maintaining state.

5.2 Active branch

On the positive active branch, $\beta(I^*)^2 = \gamma\eta/\xi - \lambda$. The Jacobian is

$$J^* = \begin{pmatrix} 2\lambda - 3\gamma\eta/\xi & \gamma \\ \eta & -\xi \end{pmatrix}. \tag{12}$$

The active-branch trace and determinant are

$$T^* = \mathrm{Tr}(J^*) = 2\lambda - 3\gamma\eta/\xi - \xi, \tag{13}$$

$$D^* = \det(J^*) = 2(\gamma\eta - \lambda\xi), \tag{14}$$

$$\eta_c = \xi\lambda/\gamma. \tag{15}$$

When $R > 1$, $D^* > 0$. The trace is also negative: since $R > 1$ implies $\gamma\eta/\xi > \lambda$, we have $3\gamma\eta/\xi > 3\lambda$, so

$$T^* < 2\lambda - 3\lambda - \xi = -\lambda - \xi < 0. \tag{16}$$

Thus the positive active branch is stable whenever it exists. The eigenvalues are

$$\mu_{1,2} = \frac{T^* \pm \sqrt{(T^*)^2 - 4D^*}}{2}. \tag{17}$$

For positive parameters in the minimal ABE, the discriminant is nonnegative on the active branch. Thus the minimal model predicts stable real relaxation rather than generic underdamped oscillation. This is not a weakness; it is a constraint. If an empirical system shows persistent oscillatory recovery, the minimal ABE is incomplete for that system and must be extended with delayed feedback, inhibitory coupling, or an additional inertial or memory state.

The robust dynamic claim is therefore stability and relaxation to the active branch, not a universal oscillation law. The robust scaling claim is the active-state amplitude law in Eq. (6).

A quantitative consequence of Eqs. (13)–(14) is critical slowing as the threshold is approached from above. The active-branch determinant satisfies

$$D^* = 2\xi\lambda(R - 1), \tag{18}$$

so $D^* \to 0$ as $R \to 1^+$. Since the smaller-magnitude eigenvalue scales as $D^*/|T^*|$ near threshold, the dominant relaxation rate vanishes there: small perturbations decay arbitrarily slowly as the system approaches the viability transition. Absence of this critical-slowing signature in a calibrated system would be inconsistent with a stable active branch born at the predicted threshold.

6 Dimensionless interpretation

The model can be written in dimensionless form to expose its control logic. After rescaling time and amplitudes, the unforced dynamics can be represented qualitatively as

$$\frac{di}{d\tau} = a(s - i - i^3), \tag{19}$$

$$\frac{ds}{d\tau} = b(Ri - s), \tag{20}$$

where a and b are positive dimensionless response rates. In this form, R is the main control number. $R < 1$ means dissipative and regulatory losses dominate entropy-supported activity. $R > 1$ means adaptive drive is strong enough to support a bounded active boundary.

The analogy to reproduction numbers in epidemiology is useful only at the level of threshold logic: a dimensionless control number crosses one, changing the qualitative behavior of the system. The ABE does not import the microscopic assumptions of epidemic models.

7 Falsifiable signatures

The ABE makes three classes of falsifiable predictions. These are model signatures, not domain-specific numerical forecasts until the parameters are calibrated in a given system.

1. **Threshold signature.** Systems should transition from externally supported activity to self-maintaining activity as R crosses one. The variables γ, η, ξ, and λ must be estimated from the target system or constrained by independent measurements.

2. **Amplitude signature.** Above threshold, the active information amplitude should obey $I^* \propto (R-1)^{1/2}$ when cubic saturation is the leading nonlinearity. Equivalently, $(I^*)^2$ should be linear in $\eta - \eta_c$ for fixed γ, ξ, λ, and β.

3. **Relaxation signature.** Under the minimal positive-coupling model, calibrated systems should relax through stable real modes toward the appropriate branch. Strong underdamped oscillation would not confirm the minimal ABE; it would indicate the need for a delayed, inhibitory, or higher-dimensional extension.

These are simple enough to be wrong. That is a strength. The ABE is useful only if its threshold and scaling laws survive contact with measured or simulated systems.

8 Candidate calibration domains

The most immediate calibration domain is adaptive computational governance, where queue pressure, latency drift, rejection rates, routing decisions, and policy outcomes can be logged and replayed. In that setting, the variable I may represent effective policy-bearing information or organized control state, S may represent operational complexity or pressure, and $\Phi(t)$ may represent incoming workload. A worked synthetic example in this style is given in Sec. 11.

Neural, biological, economic, and organizational systems may also provide calibration targets, but those applications require separate parameter estimation and should not be treated as solved by the present paper. A neural application would need defensible proxies for stored synaptic information and firing-pattern entropy. An organizational application would need proxies for institutional knowledge, turnover, procedural complexity, and innovation or learning rate. Without such calibration, the ABE remains a hypothesis-generating model, not a domain-specific theory. This paper therefore treats cross-domain use as a research program, not as proof of universal applicability.

9 Stochastic pulse realization

The deterministic ABE can be interpreted as a mean-field law. A stochastic implementation can realize the same qualitative behavior through event pulses and adaptive clustering. The point-process view is useful for computational systems because it allows a boundary law to be tested with replayable synthetic event streams before being fit to real telemetry.

One natural realization uses three coupled model families.

- **Poisson pulse emitters:** independent or conditionally independent event sources produce raw stochastic pulse timing.
- **Hawkes-style clustering:** pulse arrivals alter future pulse intensity, producing burst structure and event clustering rather than flat randomness [8].
- **Markov damping:** a finite-state process regulates transitions between calm, active, burst, recovery, and saturated regimes, preventing uncontrolled excitation [9, 10].

In short: Poisson emitters provide raw pulse timing; Hawkes-style dynamics shape burst structure; Markov damping stabilizes transitions between burst cycles. The deterministic ABE is then interpreted as the coarse-grained boundary behavior produced by many such microscopic pulse events.

A full derivation of the deterministic ABE as a mean-field limit of a Poisson–Hawkes–Markov construction is left to future work. The bridge is plausible because mean-intensity equations for self-exciting point processes are well established [11, 12], but it is not claimed as completed here.

10 Numerical verification of analytic claims

A minimal validation suite was run against the normalized equations. The goal of this section is not to validate any empirical domain. The goal is to verify that the simulations, algebra, and stability claims all use the same model. The checks evaluate the equilibrium branch, stability conditions, amplitude scaling, active-branch stability relation, and critical-slowing behavior near the viability transition. In all cases, the results match the analytical predictions.

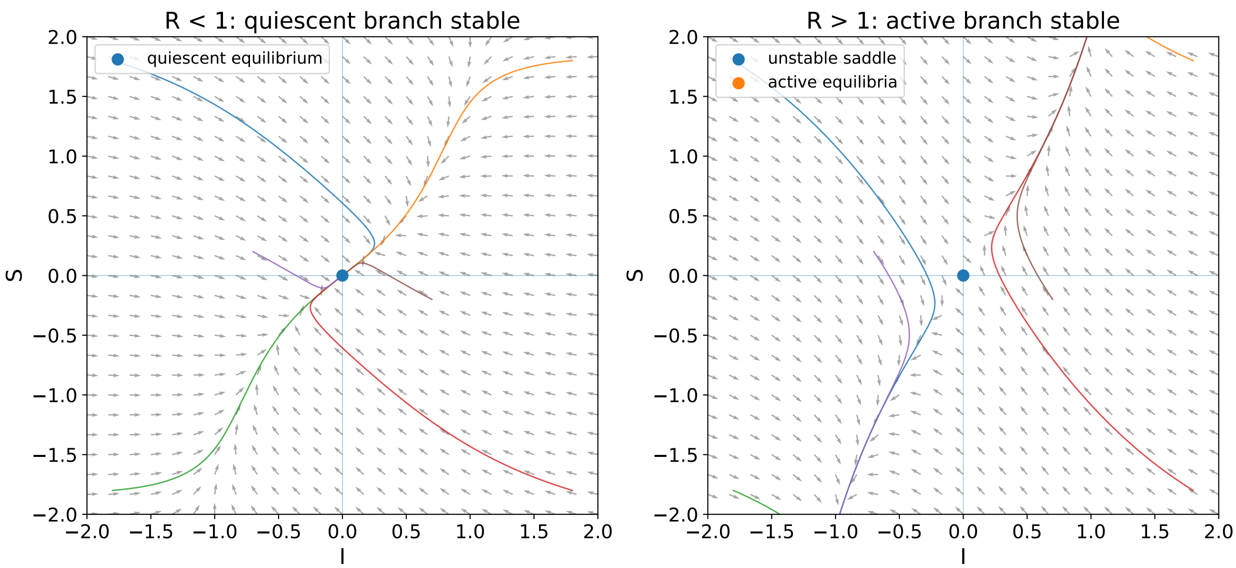

Figure 1: Phase-portrait comparison. Below threshold ($R < 1$, left), trajectories relax to the quiescent branch at the origin. Above threshold ($R > 1$, right), trajectories relax to one of the two active equilibria, while the origin becomes an unstable saddle. Equilibria are explicitly marked. This figure illustrates the qualitative state-space transition rather than empirical calibration.

A fourth numerical check verifies the critical-slowing prediction in Sec. 5.2. For fixed $\gamma = \xi = \lambda = \beta = 1$, we swept R above threshold and computed the dominant relaxation time from the smallest-magnitude real eigenvalue of the active-branch Jacobian. As predicted, the relaxation rate tends to zero as $R \to 1^+$, and the corresponding relaxation time increases sharply near the viability threshold. The result is shown in Figure 5. These figures do not replace empirical validation. They remove a more basic failure mode: claiming a threshold or oscillation law that the stated equations do not actually produce.

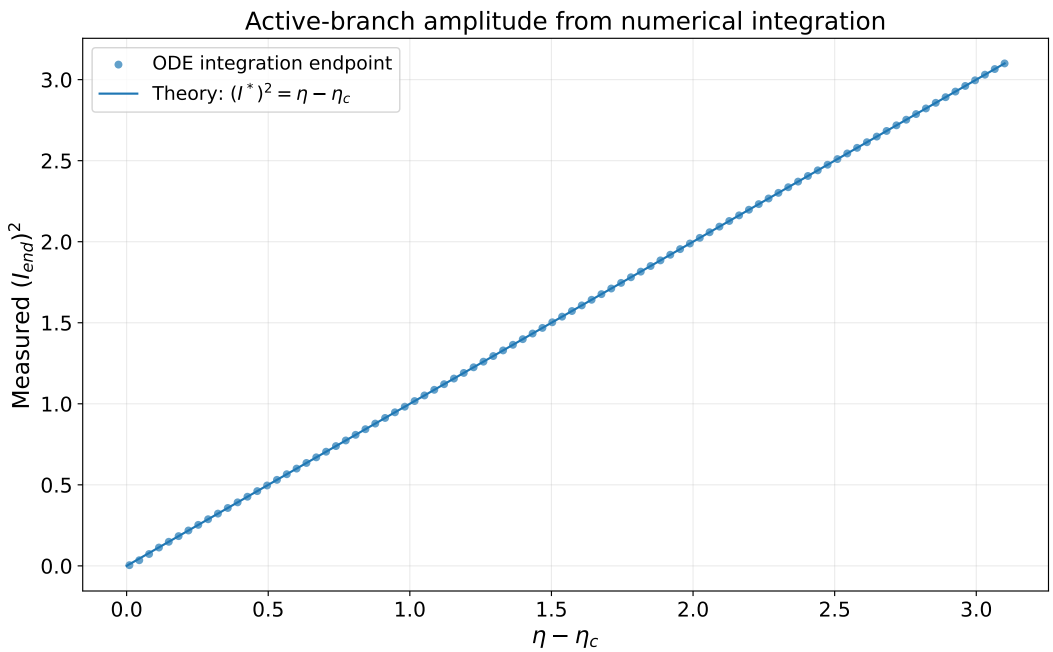

Figure 2: Active-branch amplitude scaling from ODE integration. For fixed $\gamma = \xi = \lambda = \beta = 1$, numerical trajectories were integrated from random small initial conditions and allowed to settle. The measured endpoint obeys $(I_{\text{end}})^2 \approx \eta - \eta_c$, matching the analytic prediction.

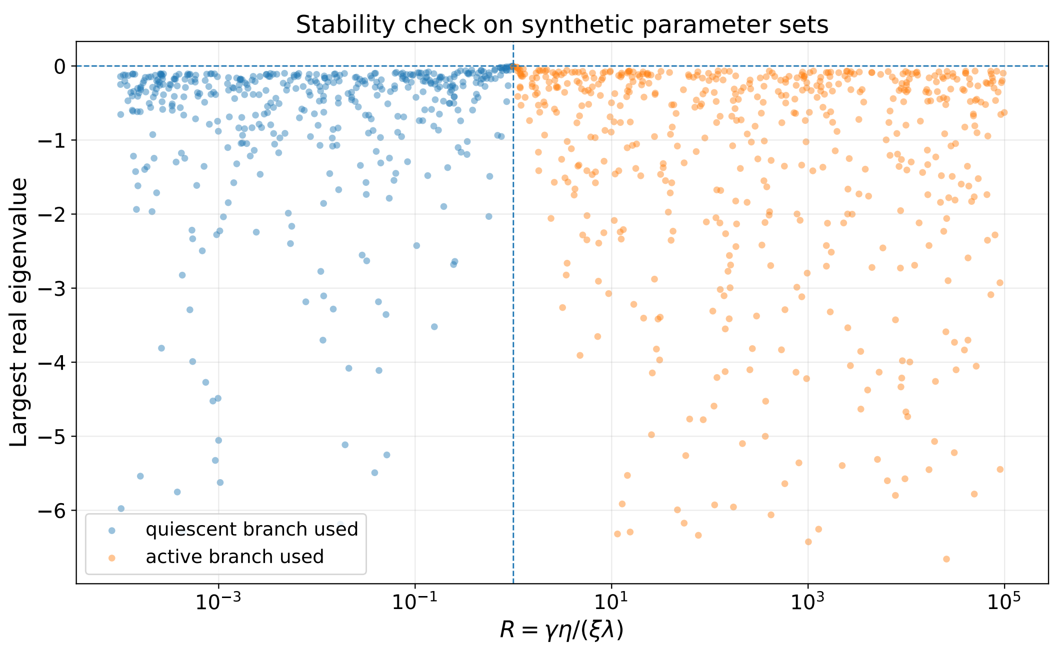

Figure 3: Stability check. Random positive parameter sets were sampled. For $R < 1$, the quiescent-branch Jacobian has negative largest real eigenvalue. For $R > 1$, the active-branch Jacobian has negative largest real eigenvalue, confirming the stability proof across the parameter space.

11 Synthetic operational demonstration

This section presents a worked demonstration in the style of an adaptive computational governor—for instance, an admission-control or backpressure system in front of a service that processes incoming work under contention. The demonstration is synthetic: the underlying generator is the ABE itself with operational measurement noise added to the equilibrium states. The point is not to validate the ABE against a real system. The point is to show what the predicted signatures look like when extracted from telemetry-shaped data, and how a practitioner would test them.

11.1 Operational mapping

In the computational-governor setting, the variables and parameters take a concrete operational meaning.

- $I(t)$: organized control state—the effective amount of policy-bearing information actively shaping admission, routing, or backpressure decisions. Practical proxies include the magnitude of the active control signal, the rate of policy-driven decisions, or the measured strength of the governor's response.
- $S(t)$: operational complexity or pressure—the current backlog of work, queue depth, or accumulated decision uncertainty. Practical proxies include normalized queue pressure, in-flight work, or rolling latency variance.
- γ: how strongly accumulated complexity drives the governor's control state. Tunable in many systems via control-loop gain.
- λ: how quickly the governor's control state decays in the absence of pressure. Tunable via decay schedules and policy half-lives.
- η: how much complexity is generated per unit of active control. Determined by workload mix and system architecture.
- ξ: how quickly accumulated complexity is dissipated by regulation, retries, or completion. Determined by downstream capacity and cleanup paths.
- β: the saturation strength preventing runaway control activity. Set by hard limits, ceilings, or rate caps.

In this mapping, the viability number $R = \gamma\eta/(\xi\lambda)$ is a dimensionless statement about whether the governor's feedback gain dominates its regulatory dissipation. $R < 1$ corresponds to a system in which control state cannot self-maintain—any active control collapses without sustained external workload. $R > 1$ corresponds to a system in which a bounded active control state can persist endogenously.

11.2 Demonstration protocol

The protocol is straightforward and applicable to any system that exposes the operational variables above:

1. Estimate γ, λ, β, and ξ from system configuration and short-time response measurements, such as control-loop gains, decay schedules, and dissipation rates.

2. Sweep η by varying the workload-to-control coupling—for example, changing admission aggressiveness, batch composition, or operator-set policy weights—while holding the other parameters fixed.

3. For each sweep point, allow the system to settle and record the equilibrium state $|I|$.

4. Plot $|I|$ against $R = \gamma\eta/(\xi\lambda)$ and $|I|^2$ against $\eta - \eta_c$. The threshold should appear at $R = 1$, and the amplitude-squared relation should be linear above threshold.

11.3 Synthetic results

Figure 4 shows the result of this protocol applied to data generated by integrating the ABE with $\gamma = \lambda = \beta = \xi = 1$, sweeping η across the threshold, and adding Gaussian measurement noise of standard deviation 0.04 to each equilibrium state to emulate real-world telemetry. Sixty parameter sweep points are shown.

Two observations are worth highlighting. First, the threshold transition at $R = 1$ is visible directly in the data without parameter fitting; it is a structural feature, not a fitted result. Second, the linear scaling of $|I|^2$ in $\eta - \eta_c$ above threshold provides a quantitative test that goes beyond the threshold itself: a system that crosses the threshold but does not show the predicted amplitude law would falsify the cubic-saturation form even while supporting the threshold structure.

This demonstration is synthetic and is therefore not evidence that any particular real system is well described by the ABE. It is evidence that the analytic predictions are observable in noisy operational-style data, and it provides a concrete protocol that other practitioners can apply to their systems.

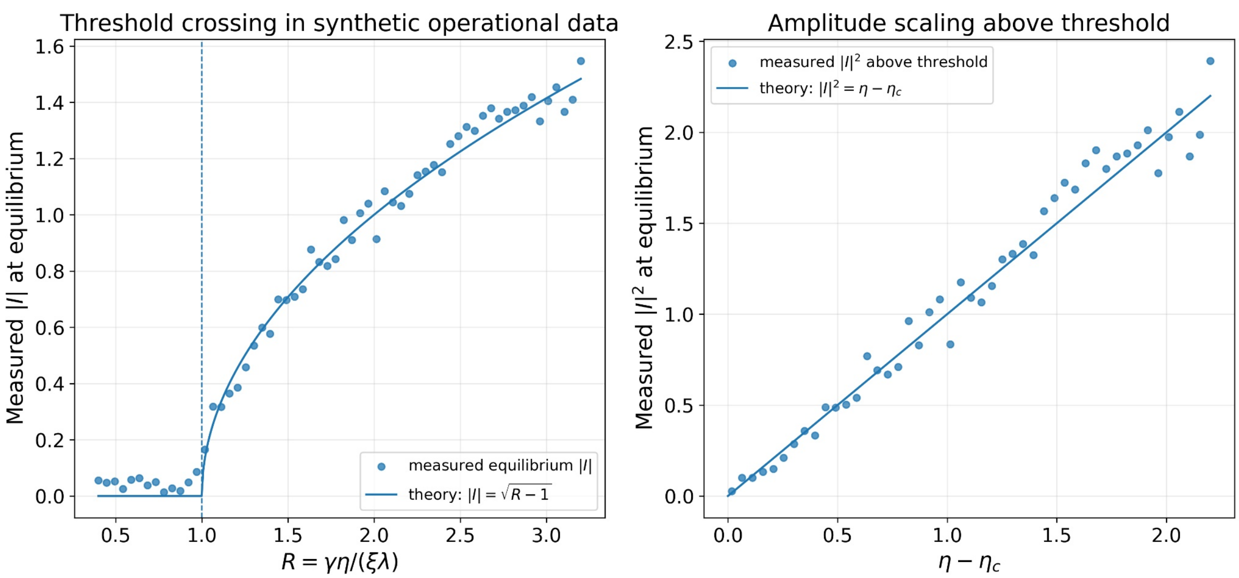

Figure 4: Synthetic operational demonstration. Left: measured equilibrium $|I|$ as a function of $R = \gamma\eta/(\xi\lambda)$. Activity is consistent with zero below $R = 1$ within the imposed measurement noise ($\sigma = 0.04$), and rises sharply at the predicted threshold. Right: above-threshold equilibrium amplitude squared, plotted against $\eta - \eta_c$. The data fall along the predicted theoretical line, with residuals consistent with the imposed measurement noise.

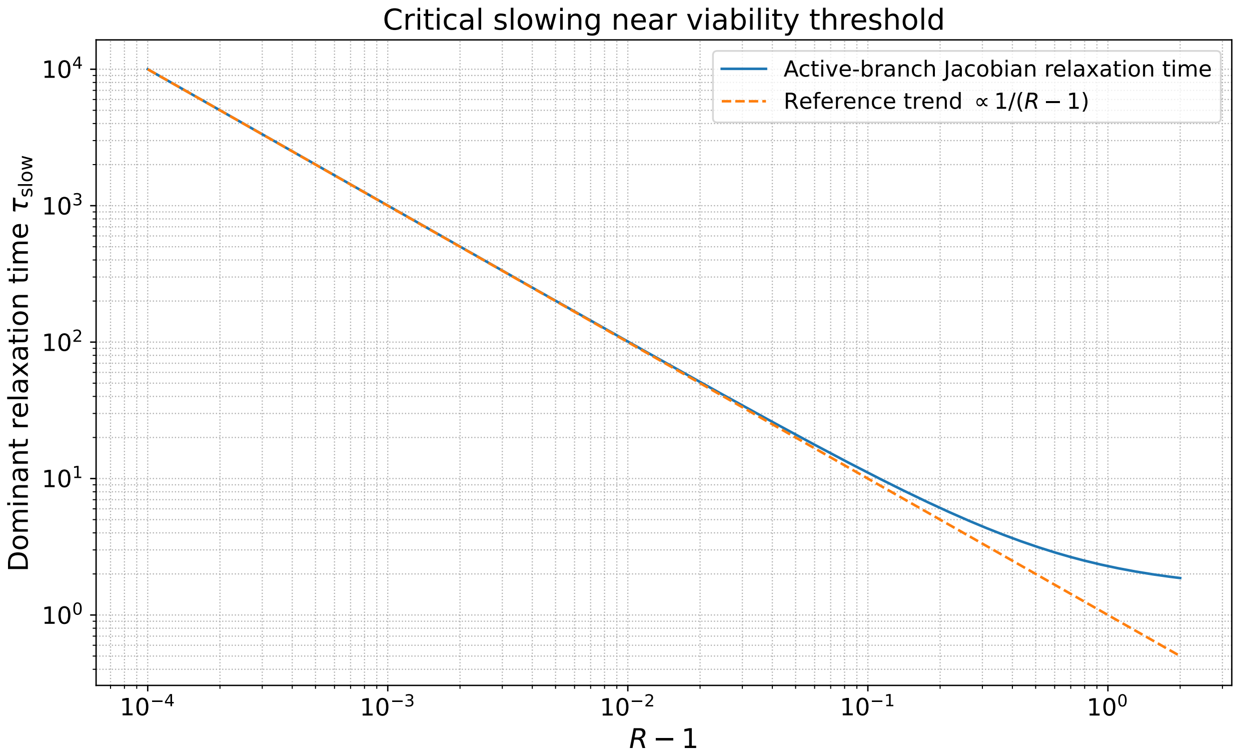

Figure 5: Critical slowing near the active-boundary threshold. For fixed $\gamma = \xi = \lambda = \beta = 1$, the dominant relaxation time is computed from the smallest-magnitude real eigenvalue of the active-branch Jacobian as R approaches 1 from above. The relaxation time increases sharply near threshold, matching the analytical prediction that the dominant relaxation rate vanishes as $D^* = 2\xi\lambda(R - 1)$ tends to zero.

12 Relationship to existing nonlinear systems

The system in Eqs. (1)–(2) shares its functional form with familiar two-variable nonlinear models, including FitzHugh–Nagumo-type relaxation systems [13, 14], cubic-saturation feedback models, and normal-form treatments near dynamical instabilities [15, 16]. In the signed-amplitude representation, the threshold has the structure of a supercritical pitchfork-type active-branch transition (see, e.g., [17] for the canonical treatment); the novelty claimed here is the active-boundary interpretation, not the normal form itself. A reader trained in nonlinear dynamics should not read the ABE as a wholly new mathematical structure.

The contribution of this paper is instead interpretive and organizational. It assigns the variables and parameters a specific active-boundary meaning: information-bearing activity, entropy or complexity, regulatory decay, entropy generation, dissipation, and saturation. It then identifies $R = \gamma\eta/(\xi\lambda)$ as a compact viability number for this interpretation, gives falsifiable threshold, amplitude, and damping signatures, and demonstrates the predictions on synthetic operational data.

Three adjacent research traditions deserve explicit mention. First, Friston and collaborators have developed a body of work on Markov blankets and active inference [18, 19] that addresses the same broad question of how systems maintain themselves through boundaries that separate internal from external states. Their framework operates through variational free-energy minimization on a generative model rather than through phenomenological ODEs, and produces a different mathematical apparatus for a structurally similar question. Second, viability theory [20] formalizes self-maintenance through set-valued dynamics and viability kernels, providing a control-theoretic framework that overlaps with the present terminology of "viability" but employs different machinery. Third, work on phenomenological model reduction in systems biology [21, 22] has shown that complex adaptive networks can often be characterized by a small number of dimensionless ratios derived from manifold-boundary approximations of microscopic kinetics, with adaptive systems sometimes reducing to a single timescale ratio. The present work differs from each of these traditions in mathematical approach: a minimal two-variable ODE with explicit threshold and amplitude predictions, falsifiable through three operational signatures, and calibrated through phenomenological parameter estimation rather than through variational inference, set-valued analysis, or microscopic reduction. The interpretive contribution is the active-boundary framing of an existing nonlinear normal form together with a viability number compact enough to be tested directly against operational data. This positioning is intentionally modest. The paper matters only if the interpretation helps connect measurable adaptive-boundary behavior to an analytically tractable model.

13 Falsification criteria

Table 1 makes the falsification structure explicit. Each row gives a class of empirical observation that, if found in a calibrated system, would either falsify the minimal ABE or indicate the type of extension required. This table is not exhaustive, but it covers the central failure modes a reviewer or practitioner would test for.

Table 1: Falsification criteria for the minimal ABE. Each observation either falsifies the minimal model or indicates the specific extension required.

Observation	**Implication for the minimal ABE**	**Candidate response**
Sustained nonzero activity in the unforced regime with R clearly < 1	Falsifies the minimal model. Activity must be either externally forced, driven by a hidden state, or supported by a different threshold law.	Re-examine forcing assumptions; consider an extended ABE with bistability or hidden state.
Quiescence persists despite R well above 1	Falsifies the minimal model in that domain. The active branch should exist and be reachable.	Investigate basin-of-attraction issues, frozen initial conditions, or missing coupling.
Active amplitude does not scale as $(I^*)^2 \propto \eta - \eta_c$ near threshold	The leading nonlinearity is not cubic. Threshold structure may still hold; the saturation law is wrong.	Replace cubic saturation with the appropriate higher-order or bounded form.
Persistent underdamped oscillation around the active branch	Falsifies the minimal positive-coupling ABE for that domain. The discriminant is forced negative by structure not in the model.	Extend with delayed feedback, inhibitory coupling, or an additional inertial state.
No critical slowing as R approaches 1 from above	Inconsistent with a stable active branch born at threshold. Either R is mis-estimated or the transition is not the one modeled.	Re-fit parameters; consider whether the empirical transition is a different bifurcation type.
Threshold appears at a value of R clearly different from 1	The chosen variable interpretation or normalization is wrong, even if the qualitative structure is right.	Revisit the operational mapping of I, S, γ, η, λ, and ξ in the target system.

These criteria are deliberately operational. A system that survives all of them is consistent with the minimal ABE in the regime tested; a system that fails one of them tells the practitioner specifically how the model needs to be modified.

14 Limitations

1. **Phenomenological status.** The ABE is not yet a microscopic derivation. Its parameters must be fit or derived for each domain.

2. **Saturation choice.** Cubic saturation is minimal and analytically convenient. Other saturation laws may change amplitudes and prefactors while preserving the threshold structure.

3. **Oscillation limits.** The minimal positive-coupling ABE does not produce a generic underdamped regime. Oscillatory systems require an extended model with delayed feedback, inhibitory coupling, or another state variable.

4. **Domain calibration.** The model does not justify precise biological, economic, physical, or computational numerical predictions without independent parameter estimation. The synthetic demonstration in Sec. 11 illustrates the protocol; it does not validate any real system.

5. **Observability.** Many systems do not directly expose both $I(t)$ and $S(t)$. Practical inference may require proxies, latent-variable estimation, or Bayesian fitting.

6. **Noise.** Stochastic forcing may shift apparent thresholds, induce switching, or blur the active transition near $R = 1$.

15 Conclusion

The Adaptive Boundary Equation proposes a minimal active-boundary law for coupled information and entropy dynamics. From Eqs. (1)–(2) follows a clear viability threshold: $\eta > \xi\lambda/\gamma$, equivalently $R = \gamma\eta/(\xi\lambda) > 1$. Below threshold, unforced boundary activity is not self-maintaining. Above threshold, a stable positive active branch exists. The active branch is stable whenever it exists. In the minimal positive-coupling form, relaxation is non-oscillatory; oscillatory domains require an extended ABE. The predicted threshold transition and amplitude scaling are visible in synthetic operational telemetry, the critical-slowing trend is verified numerically, and explicit falsification criteria are provided.

The model matters if it remains disciplined. Its value is not that it explains everything. Its value is that it gives adaptive-boundary theory a compact mathematical target that can be simulated, calibrated, falsified, and extended.

Postscript: a long-range direction

A natural long-range direction is horizon-like information–entropy relaxation, where the active-boundary interpretation becomes literal rather than metaphorical. The shared structural feature is a boundary that retains information while exchanging entropy with its environment under regulatory constraint—a description that has a formal analogy in black-hole horizon physics, while requiring an entirely separate gravitational derivation against the relevant perturbation theory and lying outside the scope of the present phenomenological treatment. We note the structural resonance only to flag it as a research direction worth separate inquiry.

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Data and code availability

A reproducibility archive accompanying this paper provides all scripts required to reproduce Figs. 1–5 and to verify the equilibrium, stability, active-branch, amplitude-scaling, and critical-slowing claims numerically, together with the LaTeX source. The archive is available at the project GitHub repository; an archived snapshot is registered under DOI 10.5281/zenodo.20090457.

Competing interests

The author is affiliated with Cripsis Logic, which is developing commercial systems related to adaptive computational control and information-governance architectures.

A Complete active-branch derivation

Starting from the unforced normalized ABE,

$$\frac{dI}{dt} = \gamma S - \lambda I - \beta I^3, \qquad \frac{dS}{dt} = \eta I - \xi S,$$

at equilibrium $dS/dt = 0$ implies $S^* = (\eta/\xi)I^*$. Substitution into the I equation yields

$$0 = (\gamma\eta/\xi - \lambda)I^* - \beta(I^*)^3.$$

Therefore $I^* = 0$ or $(I^*)^2 = (\gamma\eta/\xi - \lambda)/\beta$. The positive branch exists only if $\gamma\eta/\xi > \lambda$, equivalent to $\eta > \xi\lambda/\gamma$.

The Jacobian of the unforced system is

$$J(I, S) = \begin{pmatrix} -\lambda - 3\beta I^2 & \gamma \\ \eta & -\xi \end{pmatrix}.$$

At the quiescent state, this becomes Eq. (9). At the active branch, $\beta(I^*)^2 = \gamma\eta/\xi - \lambda$, so

$$-\lambda - 3\beta(I^*)^2 = 2\lambda - 3\gamma\eta/\xi,$$

which gives Eq. (12). The determinant is

$$\begin{aligned} D^* &= (2\lambda - 3\gamma\eta/\xi)(-\xi) - \gamma\eta \\ &= -2\lambda\xi + 3\gamma\eta - \gamma\eta \\ &= 2(\gamma\eta - \lambda\xi). \end{aligned}$$

For $\eta > \xi\lambda/\gamma$, $D^* > 0$. The trace satisfies $T^* = 2\lambda - 3\gamma\eta/\xi - \xi$. Since $\eta > \xi\lambda/\gamma$ implies $\gamma\eta/\xi > \lambda$,

$$T^* < 2\lambda - 3\lambda - \xi = -\lambda - \xi < 0.$$

Therefore the positive active branch is stable whenever it exists.

B Minimal simulation pseudocode

```
for each parameter set gamma, lambda, beta, eta, xi > 0:
    eta_c = xi * lambda / gamma
    R = eta / eta_c

    if R <= 1:
        assert no positive active branch exists
        J0 = [[-lambda, gamma], [eta, -xi]]
        assert trace(J0) < 0 and det(J0) > 0
    else:
        I_star = sqrt((gamma*eta/xi - lambda) / beta)
        S_star = (eta / xi) * I_star
        J_star = [[(2*lambda - 3*gamma*eta/xi), gamma],
                  [eta, -xi]]
        T = trace(J_star)
        D = det(J_star)
        eig = eigenvalues(J_star)
        assert D > 0 and T < 0
        assert real_part(eig[0]) < 0
        assert real_part(eig[1]) < 0
```

This pseudocode prevents the most serious submission error: using one set of equations in the model and a different Jacobian in the proof.

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